# AKASH P

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github.com/akashp2002

## Summary

MCA graduate specializing in Generative AI and Agentic AI, with hands-on experience building production-ready AI systems using LangGraph, RAG, Model Context Protocol (MCP), LLMs, and FastAPI. Experienced in designing multi-agent workflows, stateful AI applications, and intelligent retrieval pipelines with persistent checkpointing. Strong foundation in Java and full-stack development, with a passion for building scalable, real-world AI applications.

## Education

**Tkm College of Engineering** 2024 – 2026
Master of Computer Application

**Morazha Co-Operative Arts and Science College** 2019 – 2022
Bachelor of Computer Application

## Projects

**Cloudflare Incident RAG System** https://github.com/akashp2002/CloudflareAssistant

- Developed a full-stack RAG system using Python, FastAPI, React, LlamaIndex, Qdrant, Groq, and Docker to answer questions from Cloudflare incident postmortems.
- Implemented hybrid retrieval by combining dense vector search and BM25 lexical search using Reciprocal Rank Fusion (RRF), followed by cross-encoder reranking for improved retrieval relevance.
- Built grounded answer generation with source citations and explicit refusal for insufficient context, and evaluated RAG quality using Ragas.
- Added end-to-end pipeline observability using LangSmith and containerized the FastAPI, React, and Qdrant services using Docker Compose.

**Multi-Agent Job Discovery System** https://github.com/akashp2002/careerpilot ai

- Developed an agentic AI system using Python, FastAPI, LangGraph, PostgreSQL, and MCP to automatically analyze resumes and discover relevant job opportunities from multiple job sources.
- Built a multi-agent workflow for resume parsing, job search, job description analysis, semantic matching, ranking, and explainable recommendations with Human-in-the-Loop (HITL) refinement.
- Implemented persistent workflow state using LangGraph AsyncPostgresSaver, enabling checkpointing, pause/resume, and iterative preference refinement without recomputing completed steps.
- Designed an MCP-based architecture to integrate external job sources and deliver ranked job recommendations with transparent matching explanations and user feedback loops.

**DeliverEase Web Application** https://github.com/akashp2002/DeliverEase

- Developed a Scheduled Delivery Management System using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Implemented route optimization using a Distance Matrix and Nearest Neighbor algorithm to generate efficient delivery sequences.
- Integrated interactive map visualization using Leaflet and OpenStreetMap for route display.
- Added real-time GPS tracking and turn-by-turn voice navigation to guide delivery agents.

## Skills

Frontend: React js, HTML, CSS

BackEnd: Node js, Express js, FastAPI, REST APIs

Programming Languages: Java, Python, SQL

Database: MongoDB, MySQL, Qdrant (Vector DB)

AI/GenAI: RAG Pipelines, LlamaIndex, Transformers, LangGraph, MCP, Prompt Engineering

Tools: Docker, Git, LangSmith, Ragas

## Certificates

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|---|---|
| Data Structures and Algorithms Design – NPTEL–IIT Kanpur | *2025* |
| MERN Stack Development | *2022–2023* |
| Deep Learning – DeepLearning.AI (Coursera) | *2026* |